

DATA: $\{x_i\}_{i=1}^N$, $x_i \in \{0,1\}$

DATA Likelihood:

$$P(x|\mu) = \mu^x \cdot (1-\mu)^{1-x}$$

Prior on μ :

$$P(\mu|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \mu^{\alpha-1} (1-\mu)^{\beta-1}$$

$$\textcircled{1} \mathcal{L}^0 = \left[\prod_{i=1}^N P(x_i|\mu) \right] \cdot P(\mu|\alpha, \beta)$$

$$\textcircled{2} \mathcal{L} = \ln(\mathcal{L}^0)$$

$$\textcircled{3} \frac{\partial \mathcal{L}}{\partial \mu} = 0 \Rightarrow \mu_{\text{MAP}} = \frac{\sum_{i=1}^N x_i + (\alpha - 1)}{N + (\alpha - 1) + (\beta - 1)}$$

$$\mathcal{L}^o = \left[\prod_{i=1}^N \underline{\mu}^{x_i} \underline{(1-\mu)}^{1-x_i} \right] \boxed{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha) \cdot \Gamma(\beta)}} \underline{\mu}^{\alpha-1} \underline{(1-\mu)}^{\beta-1}$$

= constant
C

$$= C \cdot \mu^{\sum_{i=1}^N x_i} (1-\mu)^{N - \sum_{i=1}^N x_i} \cdot \mu^{\alpha-1} (1-\mu)^{\beta-1}$$

$$= C \cdot \mu^{\left(\sum_{i=1}^N x_i + \alpha\right) - 1} (1-\mu)^{\left(N - \left(\sum_{i=1}^N x_i\right) + \beta\right) - 1}$$

OBSERVED DATA Likelihood.

weighted DATA Likelihood in MAP case.

Prior

$$= A \cdot \mu^{(\alpha)-1} (1-\mu)^{(\beta)-1}$$

//
constant

$$\text{DATA Likelihood} \times \text{Prior} \propto \text{Posterior}$$

Same
parametric
shape
by a scaling factor

Pseudo-Code for Online Prior Update

$t = 0$ (iteration)

1) START w/ initial guess for $\alpha^{(t)}, \beta^{(t)}$.

2) Estimate MAP solution for μ :

$$\mu_{\text{MAP}}^{(t)} = \frac{\sum_{i=1}^N x_i + \alpha^{(t)} - 1}{N + \alpha^{(t)} + \beta^{(t)} - 2}$$

3) Update the prior parameters

$$\alpha^{(t+1)} \leftarrow \alpha^{(t)} + \sum_{i=1}^N x_i$$

$$\beta^{(t+1)} \leftarrow \beta^{(t)} + N - \sum_{i=1}^N x_i$$

4) $t \leftarrow t + 1$, go back to step (2).

MLE

$$\mathcal{L}^0 = P(x|\mu)$$

MAP

$$\mathcal{L}^0 = \underbrace{P(x|\mu) \cdot P(\mu)}$$

$$= P(\mu|x) \cdot P(x)$$

Bayes' theorem.

$$\propto P(\mu|x)$$

DATA Likelihood

$$\mathcal{L}_{\text{MAP}}^0 = P(x | \mu, \sigma^2) \cdot P(\mu | \alpha, \beta) \cdot P(\sigma^2 | a, b)$$

$\uparrow$ Gaussian $\uparrow$ Exponential $\uparrow$ Gamma

$$P(\mu, \sigma^2 | \delta)$$

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DATA SET: $\{(x_i, t_i)\}_{i=1}^N$, $x_i \in \mathbb{R}$
 $t_i \in \mathbb{R}$
(REGRESSION)

Mapper fct. for
learning t
through data x : $y(x) = w_0 + w_1 \cdot x$

Parameters: $w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$

synthetic data to mimic what is
seen in the world:

$$t = \underbrace{-0.3}_{w_0} + \underbrace{0.5}_{w_1} \cdot x + \epsilon$$

$$\epsilon \sim G(0, \beta \cdot I)$$

OBSERVED DATA LIKELIHOOD for MAP:

$$\mathcal{L}^o = P(t|w) \cdot P(w)$$

if we assume M.S.E. objective function (w/
regularized) then $P(t|w)$ is Gaussian

w/ mean y . or we can

define $P(\epsilon = t - y | w) = G(0, \beta I)$

$$\mathcal{L}^0 = P(\omega|\varepsilon) \stackrel{t^{-y}}{=} P(\varepsilon''|\omega) \cdot P(\omega)$$

$$G(\mu_N, \Sigma_N) = G(0, \beta I) \cdot G(\mu_0, \frac{1}{\alpha} \cdot I)$$

Naive Bayes Classifier

It is a type of generative classification.

- 1) It uses the posterior probability for each class to make inferences/predictions.

For example, let $\{(x_i, t_i)\}_{i=1}^N$ with $x_i \in \mathbb{R}$
and $t_i \in \{0, 1\}$
binary classification.

- a) Learn data likelihood for each class, C_0 and C_1 .
we estimate its parameters with MLE
or MAP approach.

$$P(x|C_0) \text{ and } P(x|C_1)$$

- b) Estimate prior probability for each class.
This is a constant value.

$$P(C_0) \text{ and } P(C_1).$$

- c) Estimate posterior to make decisions.

$$P(C_i|x) = \frac{P(x|C_i) \cdot P(C_i)}{\sum_{j=0}^1 P(x|C_j) \cdot P(C_j)}, \quad i = \{0, 1\}.$$

For test point, x_{test} , we classify it as C_0 if:

$$P(C_0 | x_{\text{test}}) > P(C_1 | x_{\text{test}}).$$

In general,

$$P(C_0 | x) \overset{\text{decide } C_0}{>} \underset{\text{decide } C_1}{<} P(C_1 | x)$$

OR
 $x \in C_i$ where $i = \arg \max_j P(C_j | x)$